Neel Kochhar

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EDUCATION

University of Calgary — Bachelor of Science in Computer Science September 2026 – Present (Expected 2030)

Relevant Coursework (Fall 2026, in progress): CPSC 231 — Introduction to Computer Science for Computer Science Majors I; MATH 265 — Calculus I; MATH 211 — Linear Algebra I

Winter 2027 (planned): CPSC 233 — Introduction to Computer Science for Computer Science Majors II; CPSC 251 — Theoretical Foundations of Computer Science I; MATH 267 — Calculus II

TECHNICAL SKILLS

Languages & Tools: Python, JavaScript, C++, SQL, HTML, Docker, Git, GitHub, Unix Shell, pytest, Raylib

Backend & LLM Engineering: FastAPI, Pydantic, SQLite, REST APIs, GitHub Apps & webhooks, Anthropic API (tool use), structured LLM output (Instructor), prompt engineering, LLM evaluation

Deep Learning & AI: PyTorch, NumPy, Transformers, Self-Attention, CNNs, RNNs, Autoencoders, Backpropagation

Mathematics: Calculus, Linear Algebra

MACHINE LEARNING AND SOFTWARE PROJECTS

Cadence — Natural-Language Calendar Agent | Python, FastAPI, Anthropic API, Docker Live Demo • GitHub

- Designed, built, and deployed (Docker, Fly.io) a full-stack calendar app where plain-English requests become events, with a JavaScript frontend (FullCalendar). Quick mode turns one sentence into a validated add/update/remove; agent mode handles multi-step requests like “cancel everything on Saturday and add a hike at 9am”.
- Wrote a tool-calling agent loop on the Anthropic SDK that feeds validation errors back to the model so it self-corrects. Designed it so the model cannot approve its own scheduling conflicts or reach another visitor’s data, with per-visitor sessions and IP rate limiting.
- Owned testing and quality: wrote 59 state-based evals that run the real model and check the final calendar (100% pass on Claude Haiku) plus 55 pytest tests. Fixed LLM date errors by giving the model a 7-day lookup table instead of asking it to do date math.

Delphi — AI Code Reviewer for New Developers | Python, FastAPI, Anthropic API, GitHub Apps Pitch • GitHub

- Built with a teammate in 26 hours at Hack the Change 2026: a GitHub App that reviews learners’ real pull requests and won’t approve one until they can explain their own code.
- Designed the Claude reviewer with Pydantic-validated structured output: a Socratic review flagging security and correctness bugs (e.g. CWE-94 code injection), hints that never give the fix, and graded questions on the diff.
- Built the FastAPI webhook pipeline: HMAC-SHA256 signature checks, an immediate response with the review run in the background, duplicate-delivery handling, and PR approval on a passing score; covered by a 99-test pytest suite.

Deep Learning Architectures from First Principles | PyTorch, Python GitHub

- Independently implemented Transformers with multi-head self-attention, CNNs, RNNs, and Autoencoders from scratch in PyTorch, validating convergence on CIFAR-10 image classification and SST-2 sentiment analysis.

Mathematical Foundations of Machine Learning | Python, NumPy GitHub

- Built a multi-layer neural network using strictly NumPy (no autograd) to solve non-linearly separable classification (XOR); derived and coded forward propagation, cross-entropy loss, and manual backpropagation.
- Authored a research paper on gradient descent analyzing loss surface topology and convergence behavior.

Interactive Pendulum Physics Engine & Simulation | C++, Raylib GitHub

- Built a real-time 2D pendulum simulation in C++ modelling gravity and damping, using delta-time frame scaling, polar-to-Cartesian mapping, an object-oriented design, and interactive mouse-drag.

LEADERSHIP & ATHLETIC ACHIEVEMENT

Assistant Karate Instructor & Dojo Volunteer 2023 – 2026

- Mentored children through 600+ volunteer hours teaching technique, conditioning, and kumite tactics; worked with head instructors on dojo operations and event organization.

Team Alberta Athlete, WKF Kumite (U21 −60 kg) 2023 – Present

- Compete in high-performance WKF kumite for Team Alberta, training daily alongside full-time studies.